

Radio

Story Context

Inside the house, the sheriff discovers a **still-active radio**.

The signal is distorted... repeating endlessly:

... --- ...

A faint voice whispers:

“same key... same door...”

Something is hidden in the transmission.

Objective

Decode the radio transmission and recover the hidden flag.

Files Provided

Description.txt
transmission.enc

Step 1 — Read the Description

```
cat Description.txt
```

Important clues:

... --- ... → SOS (Morse Code)
same key... same door...

Meaning:

- There is a hidden message (signal-based)
 - Password may be reused from previous challenge
-

Step 2 — Inspect the Transmission

```
cat transmission.enc
```

Output looks like:

```
UEsDBAoACQAAA...  
...  
THM{radio_is_broken}
```

Observation

- First part → **Base64 encoded data**
- Last line → **THM{radio_is_broken}**

This is a **fake flag (decoy)**

Step 3 — Identify Encoding

The long string:

UEsDB...

This is clearly **Base64**

Step 4 — Decode (Handling Error)

```
base64 -d transmission.enc > output.zip
```

You got:

```
base64: invalid input
```

Why?

Because:

```
THM{radio_is_broken}
```

is **not valid Base64** → causes error

Step 5 — Proper Decode Method

Option A (Clean input manually)

```
head -n -1 transmission.enc > clean.enc
```

```
base64 -d clean.enc > output.zip
```

Option B (Ignore errors — easier)

```
base64 -d transmission.enc > output.zip 2>/dev/null
```

You already did this ✓ (file still created)

Step 6 — Extract ZIP

unzip output.zip

It asks for password

Step 7 — Use Password

From hint:

same key... same door...

Password reused from Challenge 1:

shadowwalls

Step 8 — Get the Flag

cat flag.txt

Final flag:

THM{r4d10_s1gn4l_br34k_th3_c0d3}

Questions

1. What does the repeating pattern "... --- ..." translate to?

Ans: sos

2. What encoding method is used in the transmission file?

Ans: base64

3. What misleading flag is appended to the transmission?

Ans: THM{radio_is_broken}

4. What password is required to extract the hidden file?

Ans: shadowwalls

5. What is the real flag hidden in the transmission?

Ans: THM{r4d10_s1gn4l_br34k_th3_c0d3}